

US Department of Agriculture (USDA)



Statement of Work

General Information

1.0 Scope of Work

This project consists of replacing three (3) 1 inch curb stops and removing historical waterline vent features. Two (2) of the new curb stops shall "T" off the mainline (1" PE pipe) with one for isolating flow to the flush toilets and the other for isolating the flow to the shower house. The third curb stop is on the main line that isolates the hydrant at the end of the parking loop. Adjustable standpipes shall include pentagon bolt caps. The work area shall be excavated by exposing the main line and all items to be replaced and or removed. The existing vents shall be removed. A new segment of one (1") PE waterline shall be installed and then the 3 curb stops shall be furnished and installed in conformance with specifications, drawings and construction notes. The line shall be pressure tested to ensure that there are no leaks prior to backfilling. The pipe shall be bedded with washed sand, and the curb stops shall have drainage rock. Care shall be taken as not to damage existing sidewalks and or curbs.

2.0 Location

This project is located at Buffalo Gap Campground on the Dakota Prairie Grasslands, Medora Ranger District approximately 7 miles west of Medora, North Dakota, via Interstate Highway 94 in Section 9 T140N R103W in Golden Valley County, North Dakota.

Contractor Requirements

3.0 Technical Requirements / Tasks

The contractor must be licensed to perform the work as described/required. North Dakota One Call protocol (811) shall be followed.

EXCAVATION

- A. OSHA excavations standards: Meet current OSHA Safety and Health Standards for all excavation, trenching, shoring, and related work under this contract. Trench boxes, trench shields, sloping or benching systems, or suitable shoring meeting trench excavation standards of "Occupational Safety and Health Standards for the Construction Industry" (29 CFR Part 1926, Subpart P-Excavations) shall be used for excavations over 5 feet deep.
- B. Water in Excavations: Prevent surface water from flowing into excavations. If water is encountered during trench excavation, provide adequate pumping equipment and piping to keep excavation free from water. Dispose water by methods that will not impact water courses.
- C. Do not mix other excavated material with topsoil.
- D. Excavate only when materials are available on site for installation.
- E. Material to be excavated is unclassified.
- F. Excavate the depth as required for the space of specified bedding layer. If rocks over 1-inch in maximum dimension are encountered in the lower 12 inches of excavation sidewall or at the bottom of the excavation, excavate at least 4 inches below the bottom of the pipe for backfilling with Type 1 Pipe Bedding.
- G. Excavate adequate width to install and join the pipes and ball valves as specified.
- H. When soft or unstable material is encountered at the subgrade which will not uniformly support the pipe, excavate the material to the depth in accordance with the drawings and specifications and backfill to excavation subgrade elevation with Type 2 Pipe Bedding.
- I. Excavate to provide minimum cover of 6 feet over top of new HDPE water line unless otherwise shown on the drawings.
- K. Complete excavations to the valves shown in the drawings.

OVER-EXCAVATION

Unauthorized over-excavation beyond limits indicated on the drawings shall be backfilled with Type 2 Bedding Material. Backfill material and compaction of unauthorized over-excavation shall be completed at the Contractor's expense.

BACKFILLING

Prior to backfilling the line shall be tested for leaks to the satisfaction of the CO. Once the pressure test and no leaks are identified backfilling can proceed by first placing select bedding (washed sand) 4 inches deep over the new installation. Backfill remaining depth of excavation with excavated material in compacted lifts with the first lift being 2 to 3 foot in depth and compacted. Use suitable material for the first lift. Subsequent lifts shall be 12"

MEASUREMENT AND PAYMENT

Work performed under Excavation & Backfilling is incidental to work performed under Valves Installation. No separate payment will be made for work under this Section.

PART II – MATERIALS

CURB STOP QUARTER TURN CHECK AND DRAIN

Curb stop conforming to AWWA C800, quarter turn check with drain and Minneapolis top thread, stop and drain feature, pack joint for PE pipe or iron pipe threaded as necessary to fit the installation, size as shown on the drawings. Use Mueller, Ford, or approved equal.

CURB BOX EXTENSION

Cast iron extension type curb box, Minneapolis pattern, cast iron lid with brass pentagon plug, compatible with curb stop top thread and bury depth. Stationary rod extending to within 3 to 15 inches of finished grade. Provide an operating key per box to operate curb stops with the stationary rod. Use Mueller, Ford, or approved equal.

POLYETHYLENE ENCASEMENT

Wrap all direct bury cast iron ductile iron pipe and fittings including valves, valve boxes, and all other metal parts and surfaces, in polyethylene encasement.

PART III - EXECUTION

REMOVAL AND DISPOSAL OF EXISTING INSTALLATION

Existing vent standpipes, curb stops and associated mainline PE pipe designated for removal, shall be removed and disposed of in accordance with all local, state and federal statutes in an approved landfill.

CURB STOP AND CURB BOX

A Curb Stop with Drain Feature: Provide drainage at the curb stop base by placing minimum 1/2 cubic yard of 1-1/2 inch minus washed sand under and around base area. Place curb stop base on 8-inch x 8-inch x 16-inch concrete block. Extend washed rock to at least 1 foot on all sides from the base of the curb stop to at least 6 inches above the drain opening. Cover washed sand with geotextile fabric, minimum 6 oz./square yard.

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